Satellite-Derived Aerosol Optical Depth Fusion Combining Active and Passive Remote Sensing Based on Bayesian Maximum Entropy

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***Abstract*— Satellite-derived aerosol optical depth (AOD) is an important parameter for studies related to atmospheric environment, climate change, and biogeochemical cycle. Unfortunately, the relatively high data missing ratio of satellite-derived AOD limits the atmosphere-related research and applications to a certain extent. Accordingly, numerous AOD fusion algorithms have been proposed in recent years. However, most of these algorithms focused on merging AOD products from multiple passive sensors, which cannot complementarily recover the AOD missing values due to cloud obscuration and the misidentification between optically thin cloud and aerosols. In order to address these issues, a spatiotemporal AOD fusion framework combining active and passive remote sensing based on Bayesian maximum entropy methodology (AP-BME) is developed to provide satellite-derived AOD data sets with high spatial coverage and good accuracy in large scale. The results demonstrate that AP-BME fusion significantly improves the spatial coverage of AOD, from an averaged spatial completeness of 27.9%–92.8% in the study areas, in which the spatial coverage improves from 91.1% to 92.8% when introducing Cloud–Aerosol Lidar with Orthogonal Polarization (CALIOP) AOD data sets into the fusion process. Meanwhile, the accuracy of recovered AOD nearly maintains that of the original satellite AOD products, based on evaluation against ground-based Aerosol Robotic Network (AERONET) AOD. Moreover, the efficacy of the active sensor in AOD fusion is discussed through overall accuracy comparison and two case analyses, which shows that the provision of key aerosol information by the active sensor on haze condition or under thin cloud is important for not only restoring the real haze situations but also avoiding AOD overestimation caused by cloud optical depth (COD) contamination in AOD fusion results.**

***Index Terms*— Active–passive fusion, aerosol optical depth (AOD) recovery, Bayesian maximum entropy, large scale, Moderate Resolution Imaging Spectroradiometer (MODIS).**

Manuscript received July 11, 2020; revised November 21, 2020; accepted January 6, 2021. Date of publication January 28, 2021; date of current version December 3, 2021. This work was supported in part by the National Key Research and Development Program of China under Grant 2017YFC0212601 and Grant 2016YFC0200900; in part by the China Postdoctoral Science Foundation under Grant 2020M682486; and in part by the National Natural Science Foundation of China under Grant 41571344. *(Corresponding author: Tianhao Zhang.)*

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This article has supplementary material provided by the authors and color versions of one or more figures available at https://doi.org/10.1109/TGRS.2021.3051799.

Digital Object Identifier 10.1109/TGRS.2021.3051799

I. Introduction

ATMOSPHERIC aerosols, which are solid and liquid particulates generally from natural and anthropogenic sources, are a major uncertainty related to climate change, due to their direct effects by scattering and absorbing solar radiation and indirect effects by influencing the radiative properties and lifetimes of clouds [1]–[7]. Additionally, aerosol particulates have been proven to be closely associated with the human health and biogeochemical cycle [8]–[11]. Aerosol optical depth (AOD), which represents aerosol extinction effects in the vertical atmospheric column, is a widely used parameter in atmosphere-related research. Due to its spatially continuous observation, satellite-derived AOD possesses unique advantages over ground-based observations in assessing regional and global aerosol effects [12], [13]. However, the current commonly used AOD data sets, which are retrieved from satellite sensors including but not limited to Moderate Resolution Imaging Spectroradiometer (MODIS), Sea-viewing Wide Field-of-view Sensor (SeaWiFS), and Multi-angle Imaging SpectroRadiometer (MISR), hardly achieve high spatial coverage primarily due to assumptions in AOD retrieval algorithms and cloud obscurations [14]. The data missing of AOD products leads to limited information of spatiotemporal trend of

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aerosols, further affecting the atmosphere-related applications and research; thus, it is meaningful and necessary to obtain AOD products with relatively high spatiotemporal coverage to enhance the availability of data [15], [16].

Since the AOD products derived from different satellite sensors are complementary to some extent in spatial completeness and data accuracy, combining the advantages of different types of sensors could yield fused AOD products with larger coverage and better quality [17], [18]. A variety of approaches have been developed for fusing AOD products from different sensors in previous studies. One type of fusing approach provides merged AOD values only at grids where original satellite products have valid AOD values, consisting of the maximum likelihood estimation (MLE) [19], the least-squares estimation (LSE) [20], the polynomial functions [21], the optimum interpolation [22], [23], and the arithmetic and weighted average [24]. Another type of fusing approach develops geostatistical models based on Tobler's first law of geography [25] to estimate missing AOD grids via spatial autocorrelation, including the geostatistical inverse model [26], universal kriging [27], [28], and their optimized ramification [29]–[31]. Further improvement incorporates temporal autocorrelation into the AOD fusion through methods such as spatiotemporal interpolation [32] and Bayesian maximum entropy method [33], which enhance the completeness and quality of recovered AOD when large areas of original AODs are missing [34]. To our knowledge, current AOD fusions are primarily based on AOD products retrieved from passive satellite sensors, benefiting from their relatively large scanning swath which contributes to a relatively short revisit period. However, it should not be ignored that the AOD products retrieved from passive satellite sensors are significantly affected by the occurrence of cloud and haze in practical applications [35], [36]. First, it has been manifested that the spatial completeness of AOD derived by passive sensors is poor due to the cloud obscurations [37]–[39]. Second, the lack of AOD values in haze period, which is possibly caused by passive sensors misidentifying haze into cloud, fails to capture the key characteristics of haze, further tending to underestimate the local or regional atmospheric pollution level [40]. Third, the cloud screening process of passive sensors probably omits cirrus or the edge of cloud to a certain extent, where this kind of thin cloud is recognized as aerosol, causing AOD value in this condition to be higher than the real aerosol situation [41]. Thus, regarding passive sensors, it can be concluded that cloud obscuration causes the major AOD missing while misidentification between cloud and haze contributes to the uncertainty in the accuracy of AOD data sets, where fundamental improvement could be hardly achieved by merely merging multiple AOD products derived from passive satellite sensors. Satellite sensor via active measurement technique, such as Cloud–Aerosol Lidar with Orthogonal Polarization (CALIOP), is designed to provide vertical profiles of atmospheric columns, which is a complementary atmospheric measurement to that of passive technique [42]. The CALIOP could not only distinguish optically semitransparent cloud from aerosol, but also possess the ability to retrieve AOD below relatively thin cloud [43], [44].

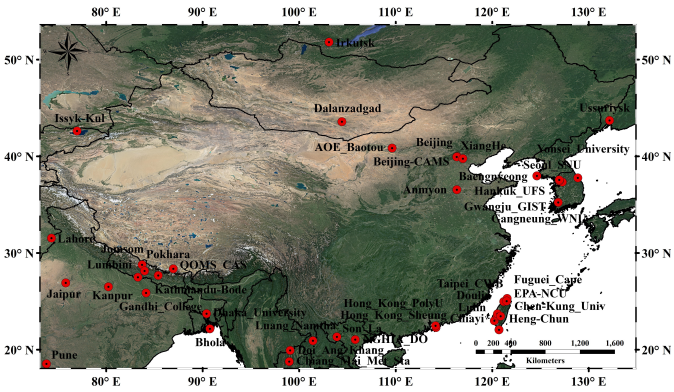

Fig. 1. Study regions and distribution of AERONET sites (indicated by red circles).

Therefore, the AOD fusions based on the combination of active and passive remote sensing could essentially improve both data accuracy and spatiotemporal coverage of AOD products.

The primary objective of this study is to develop a spatiotemporal AOD fusion framework to merge multiple satellite products from passive and active sensors based on Bayesian maximum entropy methodology. To be specific, the developed algorithm is pixel-based to estimate the AOD expectation based on the Bayesian theorem on the constraint of spatiotemporal autocorrelations in AODs. Moreover, the completeness and accuracy of AOD fusion products via AP-BME are presented and discussed to evaluate the performance of recovered AODs. Furthermore, the advantages of combining active and passive sensors for AOD retrieval are discussed through overall accuracy comparison and regional case analysis in Beijing–Tianjin–Hebei (BTH) and Yangtze River Delta (YRD) under different cloud/haze conditions.

II. Study Region and Data

A. Study Region

As shown in Fig. 1, the experimental region (73° E–135° E, 18° N–54° N) is mainly located in East Asia covering the mainland of China and its surrounding areas. It is chosen because part of regions in this area are generally characterized by high population density and atmospheric pollution level, which attracts broad attention from the public and researchers [45]. Therefore, it is essential to obtain the spatiotemporally continuous and accurate AOD as possible as it could for this area. Moreover, the sharp and complex variation of terrain in this region, which typically include the third pole, i.e., the Qinghai–Tibet Plateau, helps to evaluate the universality of AP-BME algorithm.

B. Experimental Data Sets

The data sets employed in this study consist of the AOD retrievals from MODIS and atmospheric profile data from CALIOP, which are selected to represent passive and active satellite sensors, respectively. Meanwhile, the ground-based Aerosol Robotic Network (AERONET) AOD observations, as auxiliary data, are utilized to validate and analyze the AOD fusion results. The experimental period in this study

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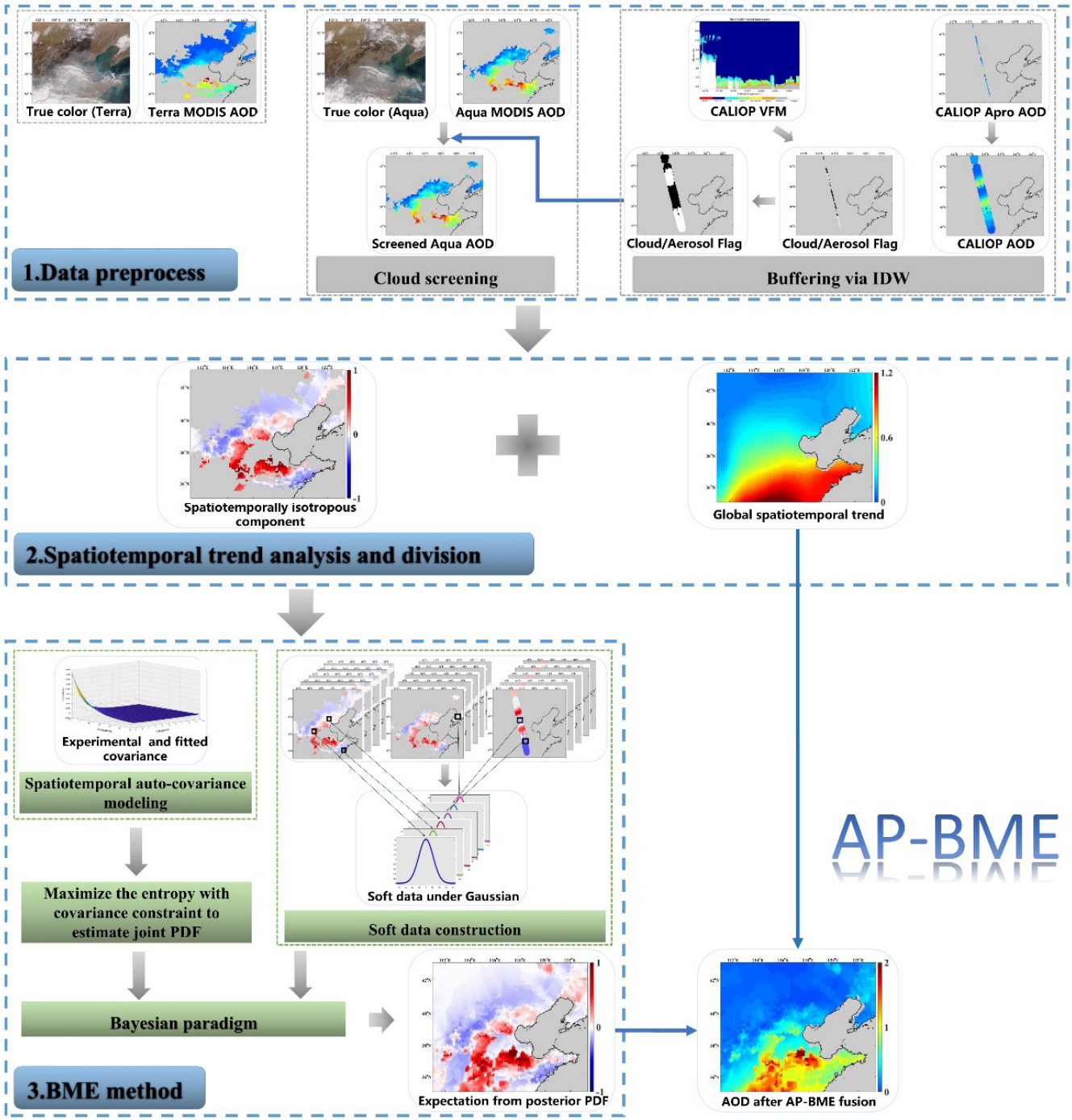

Fig. 2. Flowchart for AOD fusion combining AP-BME.

spans from January 2014 to December 2018. All the data sets including the satellite remote sensing products and ground-based observation data are briefly described in Table I and explained in detail as follows.

1) Satellite Data Sets: The MODIS instrument is onboard NASA Earth Observing System Terra and Aqua satellites, which cross through the equator at approximately 10:30 A.M. and 1:30 P.M. local time (LT), respectively. As the deep blue (DB) algorithm possesses the ability to capture aerosol properties over bright reflecting surfaces, while the dark target (DT) algorithm generally yields better AOD estimation on dense vegetated surfaces, the merged DT-DB AOD data sets derived from MODIS could achieve larger spatial coverage on land areas except for snow or ice cover [46]–[51]. As displayed in Table I, the merged DT-DB AOD data sets with 10-km spatial resolution and relatively high-quality assurance (QA) flag (QA flag = 2 or 3) are extracted from the MODIS Collection 6.1 (C6.1) Level-2 (L2) aerosol products. The MODIS C6.1 L2 aerosol products, namely, "MOD04_L2" and "MYD04_L2," are obtained from the Level-1 and Atmosphere Archive & Distribution System (LAADS, http://ladsweb.nascom.nasa.gov/).

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TABLE I
SUMMARY OF THE EXPERIMENTAL DATA SETS USED IN THE AOD FUSION FRAMEWORK

Instrument	Product	Horizontal Resolution	Collection/ Version
Terra MODIS	MOD04_L2	10 km	C6.1
Aqua MODIS	MYD04_L2	10 km	C6.1
CALIOP	CAL_LID_L2_05kmAPro	5 km	V4.20
CALIOP	CAL_LID_L2_VFM	5 km	V4.20
AERONET	AERONET Level 2.0	–	Version 3

The active sensor CALIOP is onboard the Cloud-Aerosol Lidar and Infrared Pathfinder Satellite Observation (CALIPSO), which is a member of the A-train constellation that crosses the equator at approximately 1:30 P.M. LT. The CALIOP data utilized in this study consist of two CALIPSO Version 4.2 (V4.2) Level-2 standard data products, which adopt improved algorithm especially in retrieving elevated aerosols and revising lidar ratio for few aerosol subtypes, thus possessing better performance on AOD retrieval when comparing to the previous versions [52]–[54]. One is the aerosol profile (APro) format that provides the AOD retrieval with atmospheric volume description (AVD). The AOD retrievals are screened if clouds or dense aerosol layers limit CALIOP profiling to the surface, corresponding to: 1) surface (AVD = 5) is not present at any profile range or 2) aerosol particle scattering (AVD = 3) is not detected within 250 m above the surface [42], [55]–[57]. The other is the vertical feature mask (VFM) format, which provides information on the vertical and morphological distributions of detectable features, such as aerosol, cloud, clean air, and so on. It helps to construct the data set of cloud/aerosol identification flag from CALIOP in this study, and provides references for quality control on CALIOP AOD retrievals under the clouds. The products mentioned above can be downloaded from Atmospheric Science Data Center, Hampton, VA, USA (https://eosweb.larc.nasa.gov/) and provided with 5-km horizontal spatial resolution [58], [59].

2) Ground Observation Data: AERONET AOD is the observation from a global network of calibrated sun photometers [60], which is generally employed as truth value to validate the satellite-based AOD retrievals, since the AERONET AOD is relatively accurate with an expected uncertainty of merely ±0.01 to ±0.02 [61], [62]. The AERONET Level 2.0 cloud-screened and quality-assured AOD data from 42 observation sites, which are shown in Fig. 1, are acquired from the official website (http://aeronet.gsfc.nasa.gov), and the elevation and geographic location of AERONET sites are displayed in supporting information (SI) Table I. Given that AERONET provides AOD observations for several spectral bands other than 0.55 μm, the AOD retrieval at 0.55 μm is calculated by interpolating the AODs at 0.44 and 0.87 μm based on the Ångström exponent for the 0.44–0.87-μm wavelength pair [63].

III. ACTIVE/PASSIVE-COMBINED AOD FUSION METHODOLOGY

In this article, we propose a fusion algorithm framework combining active and passive sensors to provide AOD data sets with relatively high spatiotemporal coverage and desired accuracy. As depicted by the flowchart in Fig. 2, the algorithm framework is mainly divided into four parts. The first part focuses on preprocessing of original data sets, including data integration, matchup strategy, and further cloud screening for MODIS. The second part divides the AOD data from MODIS and CALIOP into two key components, one of which called residual anomalies is the premise of modeling the spatiotemporal autocovariance. The primary purpose of the third part is to integrate the AOD residual components from different sensors via Bayesian maximum entropy in terms of both temporal and spatial characteristics, estimating the expectation of AOD spatiotemporally isotropous component. Finally, the missing AOD data are recovered by merging the expected AOD spatiotemporally isotropous component and global AOD trend component. The specific procedures of the AP-BME algorithm are described as follows.

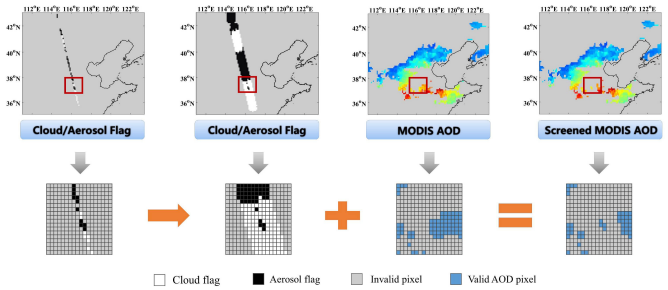

Fig. 3. Schematics of CALIOP data buffering process and cloud screening on MODIS AOD.

A. Data Preprocess Before Merging

A series of preprocessing steps are conducted on the MODIS AOD and CALIOP data sets before merging. The CALIOP data sets and MODIS AOD are first reprojected into the same coordinate system. Then the CALIOP data sets, including CALIOP AOD data and CALIOP cloud/aerosol identification flag, are resampled into 10-km spatial resolution via cubic convolution interpolation to match the MODIS DB AOD grid. As illustrated in Fig. 3, considering that the CALIOP observations have limitations in terms of the spatial coverage, the CALIOP data sets are morphologically dilated 40 km on both sides of observation track based on inverse distance weighted (IDW) [55], [64]. It should be noted that the 10-km cloud/aerosol identification grid will be set as a cloud flag, if there exists any cloud flag pixel in interpolation and dilatation calculation. Given the advantage and accuracy of CALIOP over MODIS in distinguishing cloud from aerosol, the cloud/aerosol identification flag from CALIOP is adopted to conduct a further cloud screening on the Aqua MODIS DB AOD, since the observations from these two sensors are temporally successive. Fig. 3 illustrates the schematics of further cloud screening of Aqua MODIS AOD and buffer dilation of CALIOP data sets.

B. Trend Analysis and Separation

The BME theory is constructed based on the spatiotemporal random field model (S/TRF) [65]–[67], and requires that all

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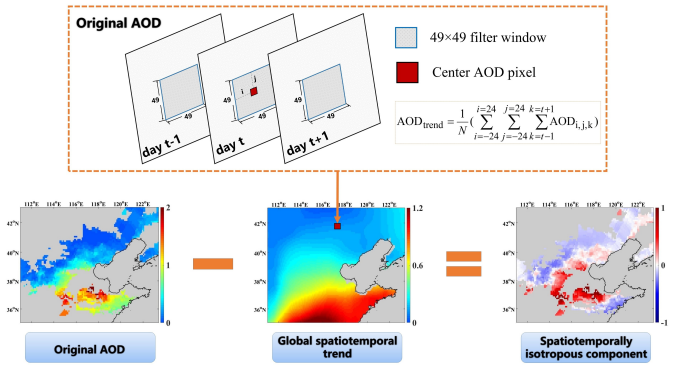

Fig. 4. Schematics of separating global spatiotemporal trend and spatiotemporally isotropous component.

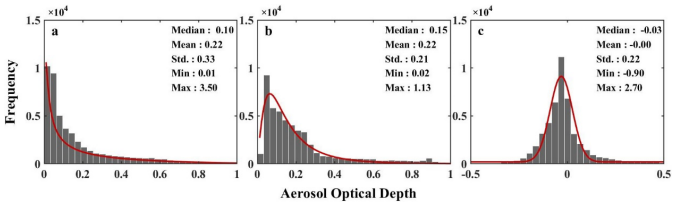

Fig. 5. Statistical descriptions of (a) original MODIS + CALIOP AOD, (b) global spatiotemporal trend, and (c) spatiotemporally isotropous component, for the entire experimental area and period.

the variables are regionalized random variables, which possess homogeneous and isotropous field. However, the original variables are generally the superposition of the global spatiotemporal trend and spatiotemporally isotropous component, as expressed as

$$\mathrm{OSD}_T(S\mathrm{variable}) = \mathrm{GSF}_T(\mathrm{variable}) + x_T(\mathrm{variable}) \quad (1)$$

where $\mathrm{OSD}_T(\mathrm{variable})$ is the original spatiotemporal distribution of the input variables, $\mathrm{GSF}_T(\mathrm{variable})$ represents the global spatiotemporal function on the day T, and $x_T(\mathrm{variable})$ is the stochastic anomalies (residuals) of the variable that can be employed to describe the spatiotemporal autocorrelation by modeling the covariance functions. Thus, before estimating the spatiotemporally autocorrelated structure of data, it is necessary to quantify and separate the global spatiotemporal component [68]. As illustrated by the schematics in Fig. 4, the global spatiotemporal trend is calculated via a spatiotemporal moving window of size 49 (pixels along-track) × 49 (pixels cross-track) × 3 (days) on the preprocessed MODIS and CALIOP AOD data sets, then the spatiotemporally isotropous component is obtained by subtracting the global spatiotemporal trend component.

From the descriptive statistics shown in Fig. 5, the original preprocessed MODIS and CALIOP AODs demonstrate a Weibull distribution [Fig. 5(a)]; the global spatiotemporal trend component is distributed in an approximately log-normal manner [Fig. 5(b)]; and the residual spatiotemporally isotropous component possesses a normal distribution [Fig. 5(c)], which meets the requirement for modeling spatiotemporal autocovariance structure.

C. Spatiotemporal Covariance Modeling

From the perspective of geostatistics, the variogram and covariance function can be used to express spatiotemporal variation and dependence of AOD, respectively, based on the assumption of second-order stationarity with constant average and variance, which are generally converse functions of separating distance in space and time [69], [70]. In this study, we adopt the nested covariance model with two exponential models proportionally summed up to adequately describe the complicated spatiotemporal variance structure [71], [72]. The nested covariance model, which is expressed as follows, is fitted by the residual isotropous AOD component using the least-squares method:

$$\begin{aligned} G\left(d = |s - s'|, \tau = |t - t'|\right) &= c_1 \exp\left(-\frac{3d}{m_{s1}}\right)\exp\left(-\frac{3\tau}{n_{t1}}\right) \\ &+ c_2 \exp\left(-\frac{3d}{m_{s2}}\right)\exp\left(-\frac{3\tau}{n_{t2}}\right). \end{aligned} \quad (2)$$

Here, d and τ are the spatial lag and temporal lag between point pairs at coordinates (s, t) and coordinates (s', t'), respectively. c_1 and c_2 are the partial sill variances of the two separate exponential models. m_{s1} and m_{s2} are the spatial ranges of fluctuation in the two exponential models. n_{t1} and n_{t2} are the temporal ranges of fluctuation in the two exponential models. The parameters for spatiotemporal covariance are modeled separately for each year, and all the annually fitting parameters are shown in SI Table II.

D. AOD Spatiotemporally Isotropous Component Estimation

The BME approach is a general and versatile framework to integrate spatiotemporal data with different extent of epistemic uncertainty [73], [74]. On the premise that spatiotemporally adjacent AOD residuals from different satellite data sets are obtainable, the expected value of missing AOD spatiotemporally isotropous component, $\overline{x}_{\mathrm{AOD}}$ on the grid (x, y) at time t, can be nonlinearly estimated from the following equation as follows:

$$\overline{x}_{\mathrm{AOD}} = \int x_{\mathrm{AOD}} f\left(x_{\mathrm{AOD}} | x_{A,1}, x_{A,2}, \ldots x_{A,n}, x_{T,1}, x_{T,2}, \ldots x_{T,n}, x_{C,1}, x_{C,2}, \ldots x_{C,n}\right) \mathrm{d}x_{\mathrm{AOD}} \quad (3)$$

where $f(x_{\mathrm{AOD}}|x_{A,1}, x_{A,2}, \ldots x_{A,n}, x_{T,1}, x_{T,2}, \ldots x_{T,n}, x_{C,1}, x_{C,2}, \ldots x_{C,n})$ is the posterior probability density function (PDF) for the adjacent AOD spatiotemporally isotropous components. $x_{A,n}$, $x_{T,n}$, and $x_{C,n}$ represent the probabilistic Gaussian soft data derived from the Aqua MODIS AOD, Terra MODIS AOD, and CALIOP AOD, respectively. The spatiotemporal range of soft data for $\overline{x}_{\mathrm{AOD}}$ depends on the spatial range and temporal range in (2), and the probabilistic Gaussian soft data are weighted as its inversed spatiotemporal distance.

According to the operational Bayesian conditionalization [68], posterior PDF $f(x_{\mathrm{AOD}}|x_{A,1}, \ldots x_{A,n}, x_{T,1}, \ldots x_{T,n}, x_{C,1}, \ldots x_{C,n})$ for each

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TABLE II

COMPARISONS BETWEEN MERGED AOD USING MODIS ONLY AND MERGED AOD USING BOTH MODIS AND CALIOP, FOR SEVERAL CITIES IN BTH ON JANUARY 8, 2017, WITH THE CLOUD FLAG AND GROUND-MONITOR PM_{10} SET AS REFERENCES

	MODIS + BME AOD	MODIS+CALIOP +BME AOD	Cloud Flag	MODIS AOD	PM_{10} (24h average: μg/m^3)
Zhangjiakou	0.2513	0.8110	T	Nan	102
Hengshui	0.3975	2.6431	T	Nan	334
Dezhou	0.8700	1.6465	F	Nan	230
Beijing	0.1950	0.1950	F	0.1950	47

estimation grid is updated from the prior PDF when Gaussian soft data are introduced, as expressed in the following equation:

$$\begin{aligned} & f\left(x_{\mathrm{AOD}} \mid x_{A,1}, \ldots x_{A,n}, x_{T,1}, \ldots x_{T,n}, x_{C,1}, \ldots x_{C,n}\right) \\ &= \frac{f\left(x_{A,1}, \ldots x_{A,n}, x_{T,1}, \ldots x_{T,n}, x_{C,1}, \ldots x_{C,n}, x_{\mathrm{AOD}}\right)}{f\left(x_{A,1}, \ldots x_{A,n}, x_{T,1}, , \ldots x_{T,n}, x_{C,1}, , \ldots x_{C,n}\right)} \\ &= \frac{f_G\left(x_{\mathrm{map}}\right)}{f\left(x_{A,1}, \ldots x_{A,n}, x_{T,1}, \ldots x_{T,n}, x_{C,1}, \ldots x_{C,n}\right)} \end{aligned} \tag{4}$$

where x_{map} is a vector of Gaussian soft data; $f(x_{A,1}, \ldots x_{A,n}, x_{T,1}, \ldots x_{T,n}, x_{C,1}, \ldots x_{C,n})$ is the prior PDF for the adjacent AOD spatiotemporally isotropous components; $f_G(x_{\mathrm{map}})$ is the joint PDF estimated by achieving maximum entropy on the constraint of general knowledge [75]–[77]. In this study, the general knowledge G is the statistical spatiotemporal covariance model for autocorrelation, which is expressed as $G(x_{\mathrm{map}})$ [61]. When predicting the probability distribution of a random event, if the entropy of the probability distribution is the largest, the probability distribution is generally the most uniform, and the predicted risk is the lowest [77], [78]. In this study, the Lagrange multiplier λ is introduced to maximize the entropy, and the joint PDF $f_{\mathrm{G}}(x_{\mathrm{map}})$ can be resolved using the following equation:

$$f_{\mathrm{G}}\left(x_{\mathrm{map}}\right) = \frac{\exp\left[\sum_{\alpha=1}^{n} \lambda_\alpha G_\alpha\left(x_{\mathrm{map}}\right)\right]}{\int \exp\left[\sum_{\alpha=1}^{n} \lambda_\alpha G_\alpha\left(x_{\mathrm{map}}\right)\right] dx_{\mathrm{map}}}. \tag{5}$$

Therefore, the expectation of AOD spatiotemporally isotropous component $\overline{x}_{\mathrm{AOD}}$ can be resolved by jointing (2)–(5). And the recovered AOD value can be obtained by adding the $\overline{x}_{\mathrm{AOD}}$ and its corresponding global spatiotemporal trend component.

IV. RESULTS AND DISCUSSION

A. Evaluation of the Completeness and Performance of Merged AOD

The primary objective of combining AODs from active and passive sensors is to improve the completeness of satellite-derived AOD products. The comparison between Fig. 6(a) and (b) illustrates that there is a dramatic improvement in the completeness of AOD. The multiyear averaged

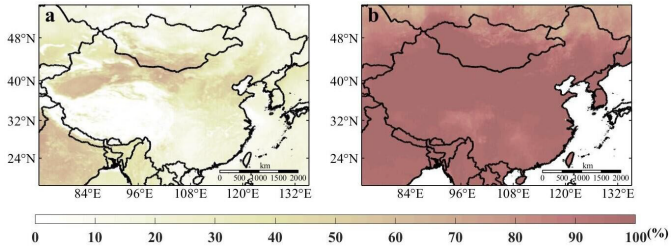

Fig. 6. Spatial distribution of multiyear averaged coverage ratio from 2014 to 2018, for (a) original MODIS/CALIOP AOD data sets and (b) merged AOD data sets.

coverage ratio of original AOD products [Fig. 6(a)] ranges from 0.3% to 75.4% with the median of 26.1%, and the spatial variation patterns indicate that the lowest coverage ratio of original AOD products appears in the plateau regions of Western China, followed by the southwestern regions and parts of high-latitude areas. The multiyear averaged coverage ratio of merged AOD data [Fig. 6(b)] ranges from 56.7% to 100% with the median of 94.9%, and the spatial distribution is analogous to that of original AOD products. When the multiyear averaged completeness of original CALIOP and MODIS AOD data is higher than 20%, the multiyear averaged spatial coverage ratio of the merged AOD after AP-BME could achieve over 80% in the whole experimental domain. Fig. 7 shows a comparison of the temporal variations in the multiyear averaged spatial completeness between the merged AOD products and original AOD products during the period of 2014–2018. As shown in Fig. 7, the spatial coverage of merged AOD after AP-BME is significantly higher than that of the original MODIS and CALIOP AOD, improving from a multiyear averaged completeness of 27.9%–92.8%, in which the spatial coverage of merged AOD via MODIS and CALIOP AOD improves from 91.1% to 92.8% when comparing to the spatial coverage of merged AOD via MODIS AOD only. Over half of days in the experimental period, the spatial coverage completeness of merged AOD products exceeds 90%. Moreover, the spatial coverage completeness of merged AOD after AP-BME is basically over 90% from March to October, whereas the completeness of merged AOD in winter is relatively lower than those in the other seasons, generally ranging from 70% to 90%. As shown in Fig. 6(b), the probable reason for the lower wintertime coverage is that the Tibet Plateau possesses relatively high altitude while the northern region is located in high latitude, where large-area

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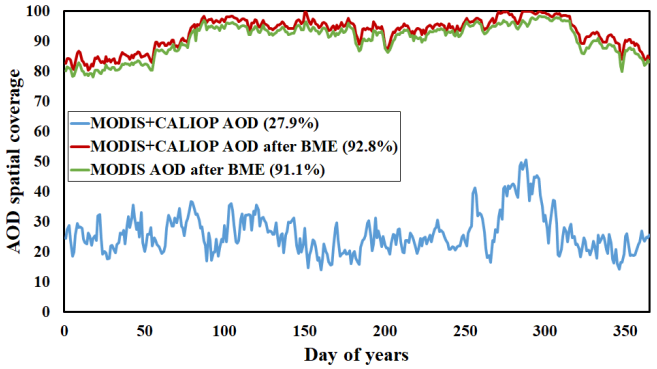

Fig. 7. Multiyear averaged daily ratio of AOD spatial coverage from 2014 to 2018, for the original MODIS/CALIOP AOD data sets (blue line), and the merged AOD data sets via MODIS/CALIOP AOD data sets (red line) and via MODIS AOD data sets only (green line).

snow/ice cover probably present for several months, especially in winter. The missing AOD could not be recovered when there exists neither spatially adjacent nor temporally adjacent original AOD products.

In order to evaluate the performance of AOD fusion process, a further discussion about the recovery details of merged AOD data is conducted through a case analysis on October 14, 2015. As shown in Fig. 8, there exist obvious missing areas in the original AOD data with spatial coverage of only about 45%, whereas the spatial coverage of merged AOD data reaches up to 100%. The red circle I in Fig. 8(b) focuses on the BTH region, which was suffering from atmospheric pollution on the selected day. It is illustrated that the fusion algorithm could recover the relatively high AODs that represent the haze situation, depending only on sparse original AOD data. Moreover, the fusion algorithm could also restore the relatively low AODs on the north and west of BTH region, located on the east edge of Loess Plateau. The similar phenomenon also appears in regions II and III, where the spatial distribution of recovered AOD corresponds reasonably to the atmospheric pollution level on different terrain structure conditions. As illustrated in Fig. 8II, the relatively higher AODs and relatively lower AODs are recovered to restore the different atmospheric situations for Sichuan Basin and Hengduan Mountain Range, respectively. For region III, the recovered AODs of Himalayas remain at a relatively low level, which escapes from the influence of relatively high AOD level in India. Overall, the AP-BME AOD fusion process performs well even when there exist sharp variations in aerosol spatial distribution due to the terrain factors, since the AOD component of global spatiotemporal trend in AP-BME framework determines the uniformity and rationality of AOD spatial distribution.

B. Assessment of the Accuracy of Merged AOD

The AERONET AOD is adopted to validate the merged AODs under different conditions and compare the accuracy of the merged AODs to that of the original MODIS/CALIOP AODs. Since the AERONET sites provide ground-level point measurements, the AOD values of both original and merged data are extracted by averaging 3×3 grids centered on the grid where the AERONET site is located. To ensure the comparability of accuracy, the matchups with an average of AERONET AOD data from 10:00 A.M. to 2:00 P.M. LT are adopted to validate the original satellite AOD products and merged AOD data. The validation results are assessed using the root-mean-square error (RMSE) [49] and the expected error (EE) envelop [79], which are defined in the following equations, respectively, where $\mathrm{AOD}_{\mathrm{origin/merge}}$ represents the original satellite AOD product or merged AOD data, and $\mathrm{AOD}_{\mathrm{AERONET}}$ represents the ground monitoring AOD data from AERONET sites:

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(\mathrm{AOD}_{\mathrm{origin/merge}} - \mathrm{AOD}_{\mathrm{AERONET}}\right)^2} \quad (6)$$

$$\mathrm{EE} = \pm 0.05 \pm 0.15 \times \mathrm{AOD}_{\mathrm{AERONET}}. \quad (7)$$

Fig. 9(a) and (b) shows the accuracy validation results of the original Terra/Aqua MODIS AOD data and CALIPSO CALIOP AOD data, respectively, which are set as references for the merged AOD validation. Fig. 9(c) demonstrates the overall accuracy of the merged AOD data over the whole experimental period and domain, while Fig. 9(d) just illustrates the accuracy of recovered AOD via merging algorithm where both original MODIS AOD and CALIOP AOD are missing. As shown in Fig. 9(d), the recovered AODs perform well, with the determine coefficient (R^2) of linear regression reaching up to 0.70, the RMSE of 0.23, and approximately 55% of the matchups falling within the EE envelop for a totally count of 7229 matchups. By comparing the accuracy between the results shown in Fig. 9(d) and (a)/(b), the recovered AOD accuracy for the regions where both MODIS and CALIOP are missing is close to the AOD accuracy for the regions where MODIS and CALIOP are available. Therefore, it is indicated that the AP-BME fusion process could not only improve the overall completeness of AOD data over 90%, but also ensure the quality of recovered AOD data.

C. Evaluating the Efficacy of Active Sensor in AOD Fusion

In order to evaluate the efficacy of CALIOP in the AOD fusion process, a control experiment is conducted to merge Terra/Aqua MODIS AOD, under the same conditions but only without CALIOP data. Fig. 9(e) and (f) shows the accuracy validation results of merged AOD where original MODIS AODs are missing, based on the data sets with and without CALIOP data, respectively. It is demonstrated that both completeness and accuracy of the recovered AOD data are further improved after introducing CALIOP data, since the total count of matchups (N) increases, the R^2 of linear regression rises from 0.67 to 0.72 with RMSE decreasing from 0.24 to 0.22, and more matchups fall within the EE envelop.

To further demonstrate the efficacy of CALIOP in the AOD fusion process, two case analyses are conducted during a haze period in BTH and under a cloudy condition in YRD, respectively. As shown in Fig. 10(a), most regions of BTH suffered from severe atmospheric pollution on January 8, 2017. Affected by the combination of cloud and haze, AOD could only be retrieved by MODIS for limited areas, which is illustrated in Fig. 10(b). Effective observations from

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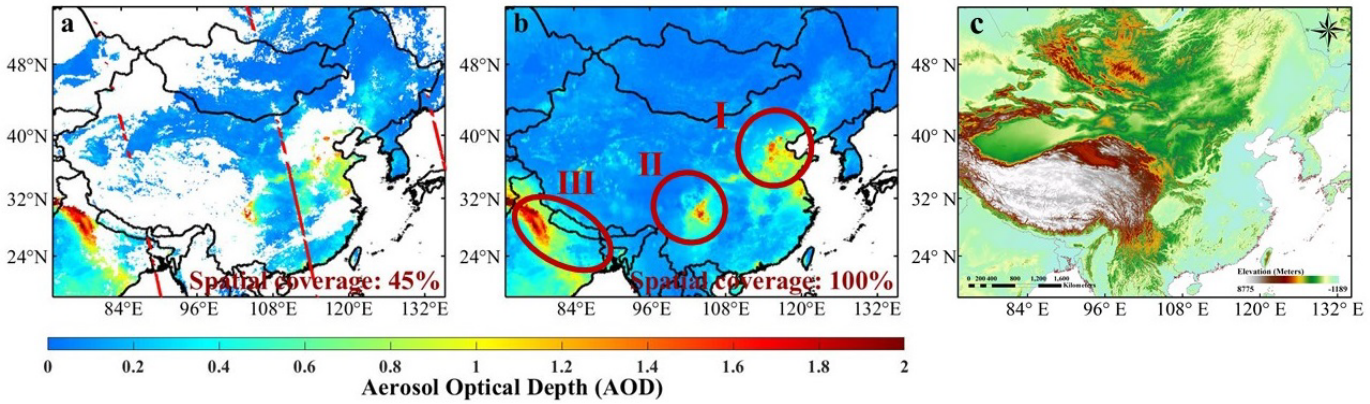

Fig. 8. Illustrations of (a) original MODIS/CALIOP AOD, (b) merged AOD after BME in the experimental regions on October 14, 2015, and (c) topography of experimental region. The red lines in the left subplot represent the track of CALIPSO nadir.

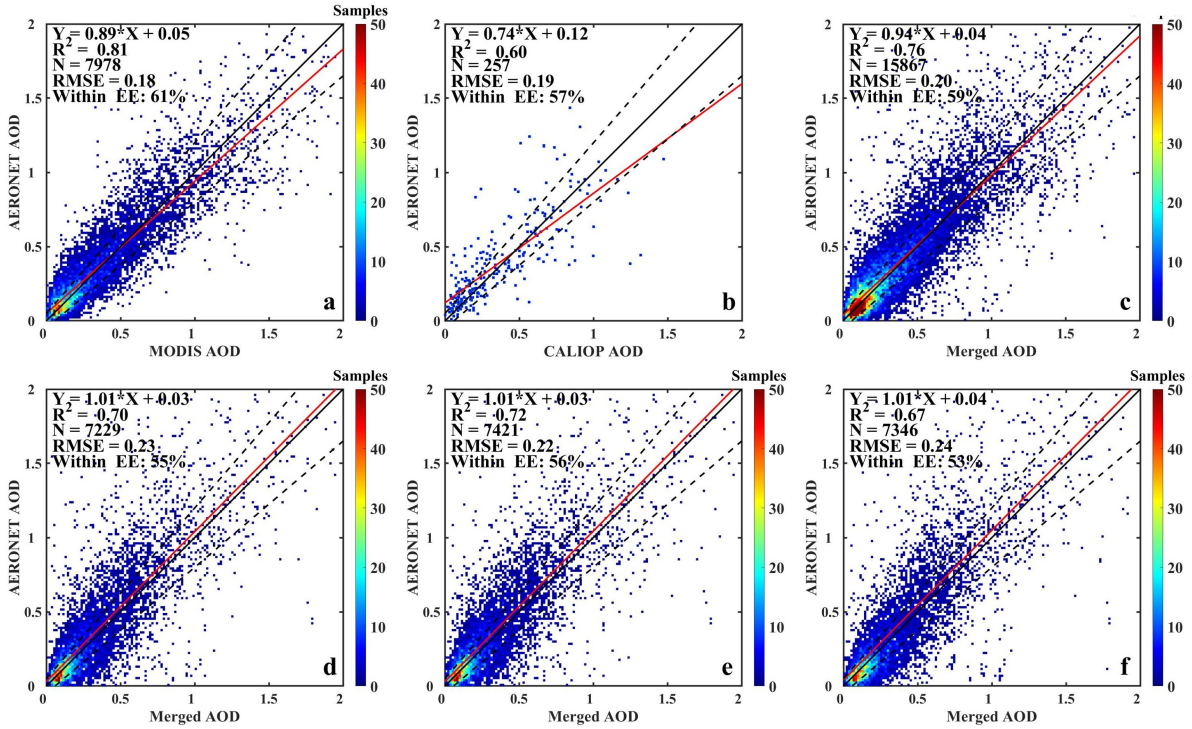

Fig. 9. Validation of the MODIS, CALIOP, and merged AOD against AERONET AOD data set. (a) Validation of the original Terra/Aqua MODIS AOD. (b) Validation of the original CALIPSO CALIOP AOD. (c) Validation of the overall merged AOD. (d) Validation of the merged AOD where both CALIOP and MODIS AODs are missing. (e) Validation of merged AOD where only MODIS AOD is missing. (f) Validation of merged AOD where only MODIS AOD is missing (the merging process did not involve CALIOP data). The black solid line represents the 1:1 reference line, the region within the black dotted lines represents the EE envelop, and the red line represents the linear regression of AOD matchups.

CALIOP, which are illustrated by red dots in Fig. 10(b), show that CALIOP could retrieve valid AOD when occurring thin cloud (northwestern BTH) and heavy haze (southeastern BTH) where MODIS failed to obtain valid AOD retrievals. Fig. 10(c) and (d) illustrates the AOD fusion results based on MODIS data and MODIS/CALIOP data, respectively. The comparison between Fig. 10(c) and (d) demonstrates that the effective CALIOP AOD retrievals on thin cloud and heavy haze conditions significantly contribute to the accuracy of AOD fusion results. As shown in Table II, in Region I, there existed atmospheric pollution to a certain extent in Zhangjiakou with daily averaged ground monitoring PM_{10} of 102 $\mu g/m^3$. In Region II, heavy haze appeared in Hengshui and Dezhou with daily averaged ground monitoring PM_{10} reaching up to 334 and 230 $\mu g/m^3$, respectively. However, the AOD fusion results using only MODIS data [Fig. 10(c)] in these three cities are at relatively low level, which is contradictory to the measurements from local ground level atmospheric monitors, indicating that there is an underestimate on the AOD fusion results using only MODIS data in these three cities, because MODIS failed to retrieve the essential AOD values about haze on cloud or heavy haze conditions. After the introduction of CALIOP data, as shown in Fig. 10(d), more accurate AODs in these three cities could be recovered to represent the atmospheric pollution situations, which is consistent with the ground monitoring PM_{10}. Admittedly,

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TABLE III

COMPARISONS BETWEEN MERGED AOD USING MODIS ONLY AND MERGED AOD USING BOTH MODIS AND CALIOP, FOR SEVERAL CITIES IN YRD ON FEBRUARY 5, 2018, WITH CLOUD FLAG AND GROUND-MONITOR PM_{10} SET AS REFERENCES

	MODIS +BME AOD	MODIS+CALIPSO +BME AOD	Cloud Flag	MODIS AOD	PM_{10} (24h average: $\mu g/m^3$)
Nanjing	0.7260	0.3010	T	Nan	91
Zhenjiang1	0.7793	0.3120	T	Nan	83
Zhenjiang2	0.8500	0.5900	T	0.8500	134
Yancheng	0.5605	0.5605	F	0.5605	129
Suzhou	0.2560	0.2560	F	0.2560	95

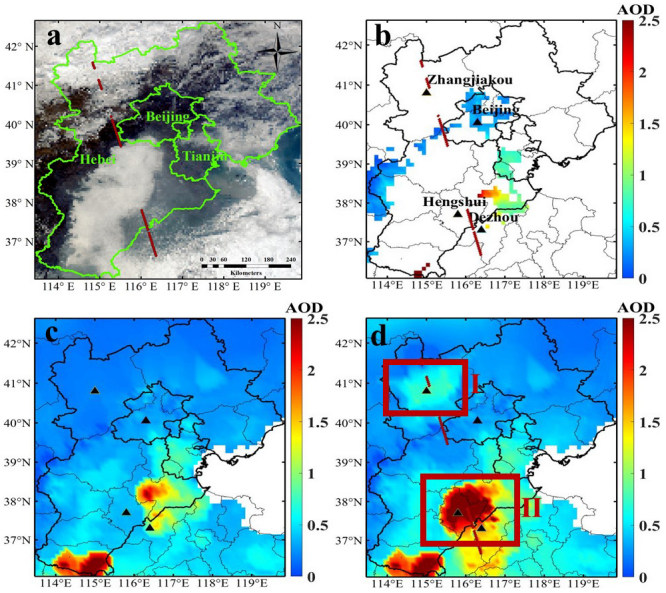

Fig. 10. Satellite image and AOD retrievals over BTH regions on January 8, 2017. (a) Satellite-derived true color image. (b) MODIS AOD. (c) Merged AOD based on MODIS AOD only. (d) Merged AOD based on MODIS AOD and CALIOP data. The red lines represent the tracks of CALIPSO nadir.

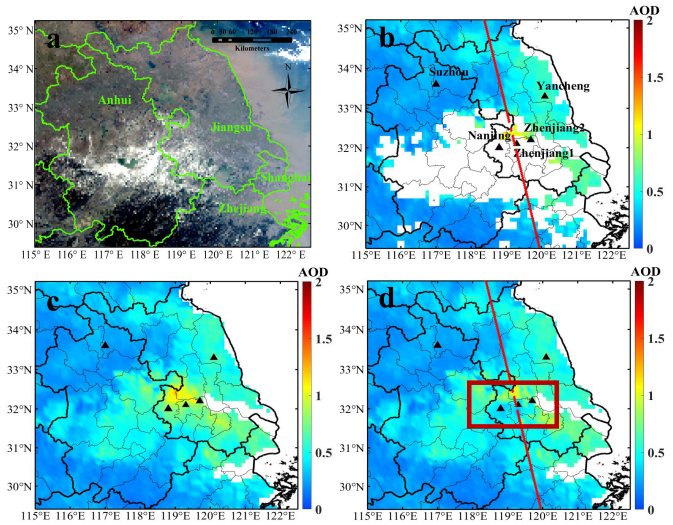

Fig. 11. Satellite image and AOD retrievals over YRD regions on February 5, 2018. (a) Satellite-derived true color image. (b) MODIS AOD. (c) Merged AOD based on MODIS AOD only. (d) Merged AOD based on MODIS AOD and CALIOP data. The red lines represent the tracks of CALIPSO nadir.

CALIOP could hardly capture effective APros when encountering thick cloud such as part of Southern BTH on that day, but the provision of key aerosol information by CALIOP under thin cloud or on haze condition is significantly useful to restore the real haze situations.

Fig. 11 illustrates a case of YRD region on February 5, 2018, when part of YRD regions were cloudy. As illustrated in Fig. 11(a) and (b), influenced by the cloud obscuration, MODIS could not retrieve valid AODs over part of YRD regions. And as shown by the red dots in Fig. 11(b), effective observations from CALIOP could be achieved for most of its scan trajectory on this thin cloud condition. Similar to the previous case, Fig. 11(c) and (d) illustrates the AOD fusion results based on MODIS data and MODIS/CALIOP data, respectively. The comparison between Fig. 11(c) and (d) demonstrates that the recovered AODs over part of the southwestern Jiangsu decrease after introducing CALIOP data, indicated by the red rectangular region in Fig. 11(d). As listed in Table III, derived from CALIOP VFM data, cloud existences are detected in cities of southwestern Jiangsu, namely Nanjing and Zhenjiang. However, some MODIS AODs on the cloud edges would be calculated and probably be considered as effective values, although these AODs are overestimated because of the contribution from cloud optical depth (COD) from the cloud edges. The overestimation of MODIS AOD on cloud edge will directly lead to the overestimation of AOD fusion results in this area and its surrounding regions. Owing to the ability to capture the information of atmospheric profile, the introduction of CALIOP data could not only precisely detect the existence of cloud edge or thin cloud, but also provide cloud-effect-free AOD to revise the overestimated MODIS AOD, further improving the quality of AOD fusion results. Thus, as shown in Table III, the AOD fusion results over Nanjing and Zhenjiang based on the combination of MODIS and CALIOP present better consistence with the ground monitoring PM_{10} concentrations than the AOD fusion results based on only MODIS, when comparing to AOD–PM_{10} relationship in Yancheng and Suzhou with no cloud obscuration on that day. Therefore, CALIOP data not only help to distinguish the cloud from aerosol more precisely, but also provide AOD value

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under thin cloud or cloud edge to avoid AOD overestimation in fusion results, which contributes to restore the real atmospheric situation under thin cloud and cloud edge conditions.

V. Conclusion

Current satellite-derived AOD products have limitation in spatial coverage and consequently it is necessary to recover the missing aerosol information. In this study, a novel AOD fusion framework combining the active and passive satellite sensors based on Bayesian maximum entropy methodology, namely AP-BME, is developed to make their respective advantages complementary to each other, especially when thin cloud or haze conditions occur. The developed algorithm is pixel-based to estimate the AOD expectation based on Bayesian theorem on the constraint of spatiotemporal autocorrelations in AODs. The results from five-year data sets in the study areas demonstrate that the spatial coverage of AOD is significantly improved after AP-BME fusion, from an averaged spatial completeness of 27.9% to 92.8%, in which the spatial coverage improved from 91.1% to 92.8% when introducing CALIOP AOD data sets into merging process. The recovered AOD performs well which corresponds reasonably to atmospheric pollution level on different terrain structure conditions. Meanwhile, the accuracy of recovered AOD against ground-based AERONET AOD shows that the R^2 of linear regression reaches up to 0.70 and approximately 55% of the matchups fall within the EE envelop for a total count of 7229 matchups, which only slightly decrease compared to original MODIS AOD products. Moreover, the efficacy of active sensor in AOD fusion is discussed through overall accuracy comparison and two regional case analyses in BTH and YRD under different cloud/haze conditions. The results show that the provision of key aerosol information by CALIOP under thin cloud or haze condition is important for not only restoring the real haze situations but also avoiding AOD overestimation caused by COD contamination in AOD fusion results.

In this study, we select MODIS and CALIOP as the representatives of passive and active satellite sensors, respectively, to evaluate the performances and advantages of AP-BME AOD fusion framework. Obviously, the proposed algorithm framework is also applicable to the AOD products derived from other satellite sensors. Admittedly, there exists limitation in the spatiotemporal coverage of CALIOP data, due to its active scanning mode. To further improve the effective scanning region of LIDAR, future work could focus on constructing an aerosol–cloud satellite active sensor network, preliminarily consisting of CALIOP, Cloud-Aerosol Transport System (CATS, onboard the International Space Station), and Atmosphere LIDAR (ATLID, onboard Earth Cloud, Aerosols and Radiation Explorer) [80] and may be extended to incorporate High Spectral Resolution Lidar (HSRL, onboard Advanced Composition Explorer) [81]. The combined use of these sensors will take full advantage of the AP-BME AOD fusion framework developed in this study. Moreover, since only CALIOP day profiles were adopted in this experiment, the AOD fusion results after AP-BME in this study represented the daytime aerosol situation. Since the CALIOP night profile generally possesses better observation performance than the day profile for its better signal-to-noise ratio, our future work will focus on yielding nighttime AOD fusion data sets with high spatial coverage by merging nighttime AOD via Visible Infrared Imaging Radiometer (VIIRS) Day/Night Band and CALIOP night profiles through AP-BME framework. And it will be more contributive to the diurnal analysis and application of aerosol when the AOD fusion data sets for daytime and nighttime are both available.

Acknowledgment

The authors would like to express their gratitude to the NASA, Washington, D.C., USA, for providing the Moderate Resolution Imaging Spectroradiometer (MODIS) and Cloud–Aerosol Lidar with Orthogonal Polarization (CALIOP) products, and the Principle Investigators for establishing and maintaining the Aerosol Robotic Network (AERONET) sites.

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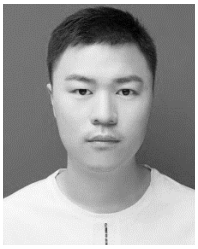

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